

RMO(INDIA) – 1996

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. The sides of a triangle are three consecutive integers and its inradius is four units. Determine the circum radius.
 2. Find all triples (a, b, c) of positive integers such that: $\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)\left(1 + \frac{1}{c}\right) = 3$.
 3. Solve for real numbers x and y : $xy^2 = 15x^2 + 17xy + 15y^2$, $x^2y = 20x^2 + 3y^2$.
 4. Suppose N is an n -digit positive integer such that:
 - (a) All the digits are distinct.
 - (b) The sum of any three consecutive digits is divisible by 5.Prove that n is at most 6. Further show that, starting with any digit, one can find a six digit number with these properties.
 5. Let ABC be a triangle and h_a be the altitude through A . Prove that $(b + c)^2 \geq a^2 + 4h_a^2$, where a, b, c are the sides of the triangle.
 6. Given any positive integer n show that there are two positive rational numbers a and b , $a \neq b$, which are not integers and which are such that $a - b, a^2 - b^2, a^3 - b^3, \dots, a^n - b^n$ are all integers.
 7. If A is a fifty element subset of the set $\{1, 2, 3, \dots, 100\}$ such that no two numbers from A add up to 100, show that A contains a square.

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